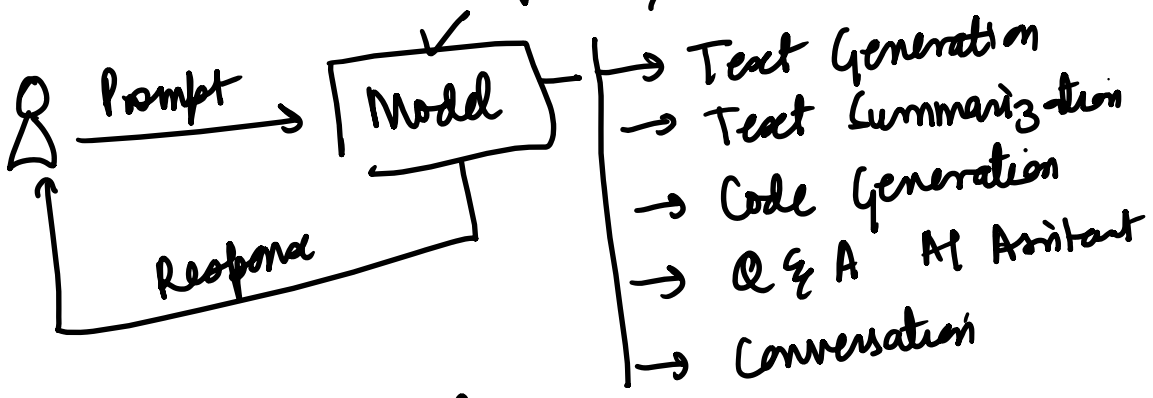
"Generative AI" ⇒ Create NEW Content – Text, Image, video, audio — Code



Traditional ML Model



Generative AI = Powered by "FM | LLM"

FM / LLM = Pretrained Model which are trained on Internet scale of data (JSON, Code, IMAGE, video)

↳ "General Domain"

LLM = Large Language Model (NLP)

"Amazon Bedrock" → 270+ Models (FM | LLM)

Input = Text, Image, video

↓

output = Text.

Basic agent Behavior

↳ "Business goal"

→ "Business goal"

↓

"Agent" interprets the goal & 'plan the actions'

↓

Data Sources | Tools / APIs | Memory

↓

Evaluate Results & deliver outcome.

Frameworks = 'Langchain, LangGraph'

= STANDS → Build-in components.

LLM = Generate a summary

GenAI Assistant = Conversation

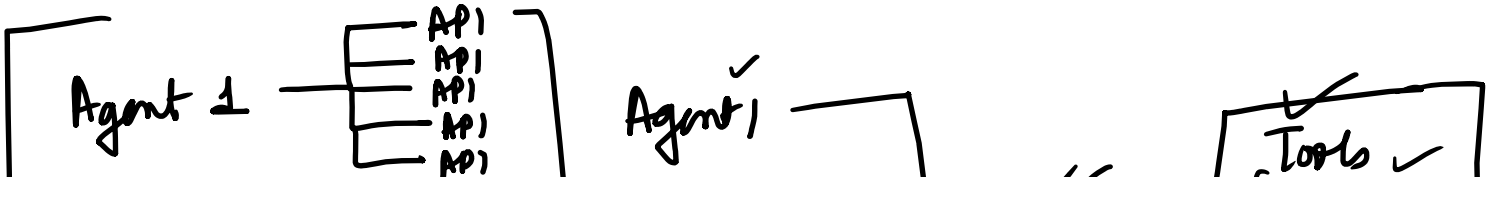
→ GenAI Agent → Retrive data, calculate, Goal, Actions

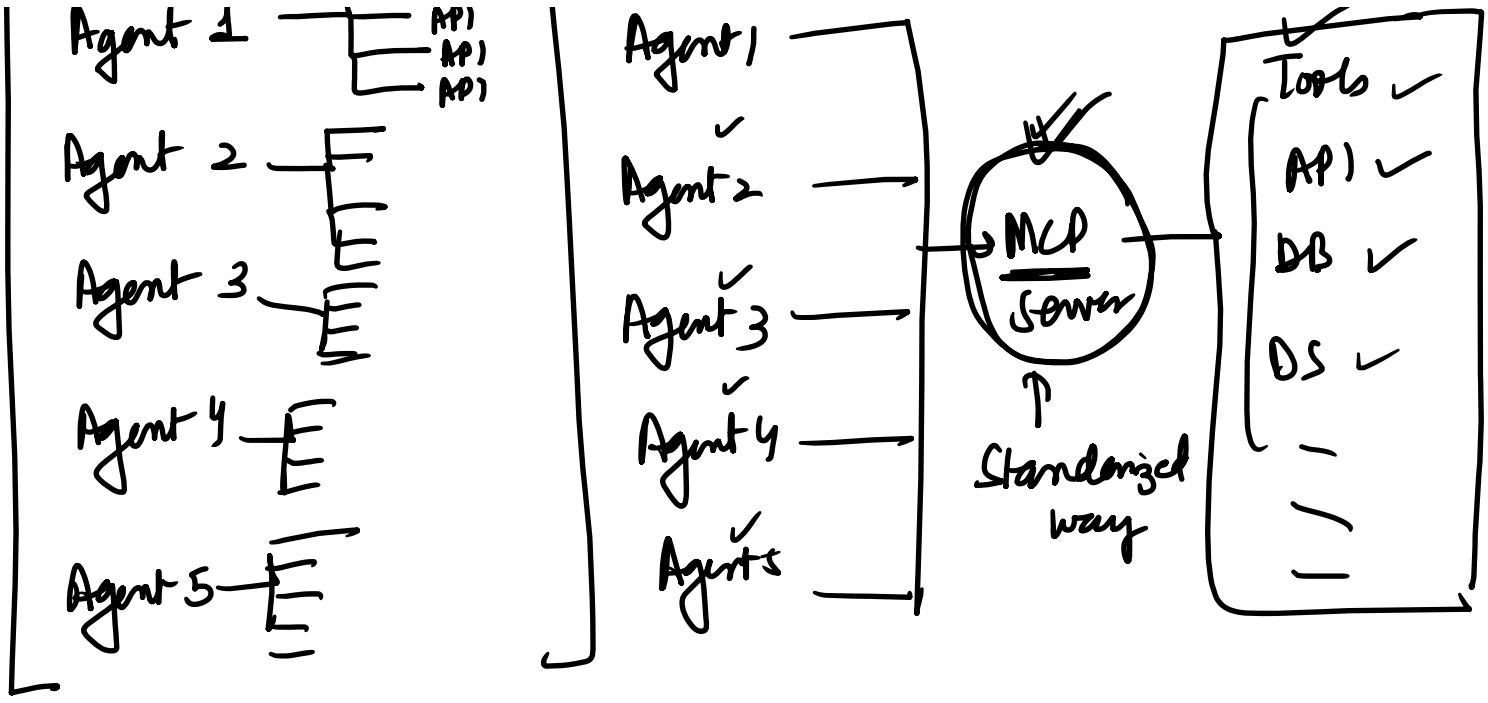
Agentic AI System → Coordinate between agents, validation, approval, monitoring, repeating.

Iterative agent behavior (Agentic loop) = goal

Observe → Reason → Act → Evaluate

Refine until the goal is reached.





Agent Type	How steps ?	Use cases
① Workflow (fixed)	Predefined steps	Repeatable, controlled Business operations
② Autonomous (Dynamic)	Dynamically selected by agent (Goal → Actions)	Open-ended investigation
③ Hybrid	Routing between fixed & dynamic	- Mixed predictable & complex
④ Multi-agent	Coordination among the agents	- Complex goals - multiple responsibilities

Workflow patterns

① Prompt chaining → What must happen first ?

Usecase : Extract fin metrics (①) ✓, validate (②) ✓, then summarize (③) ✓

② Parallelization : What can happen simultaneously ?

Usecase : Analyze five companies concurrently → summary.

Usecase: Analyze five companies ... → summary.

(3) Routing = which path should be selected?

- simple
- complex

(4) Orchestration = Who coordinates everything?

Manage an end-to-end process

Q Which pattern is responsible for selecting the legal-review path? (choose a path)

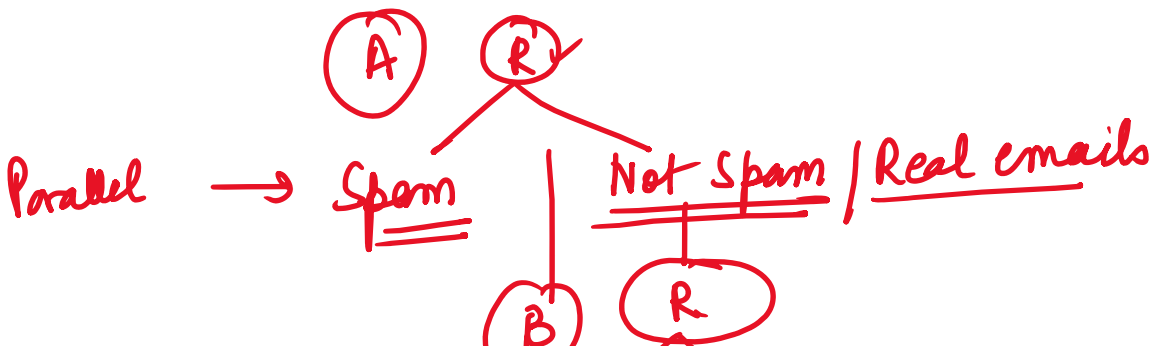
Ans: Routing

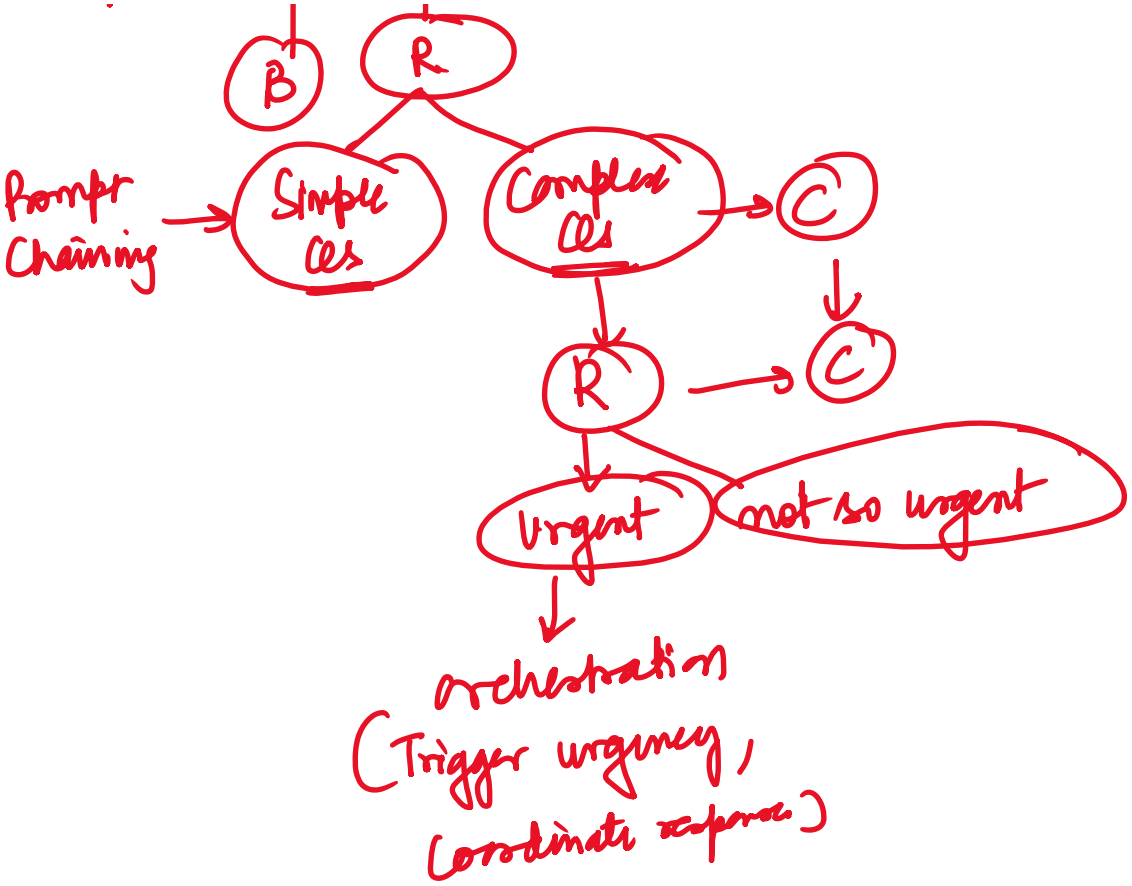
Q Which pattern coordinates the complete request?

Ans: Orchestration

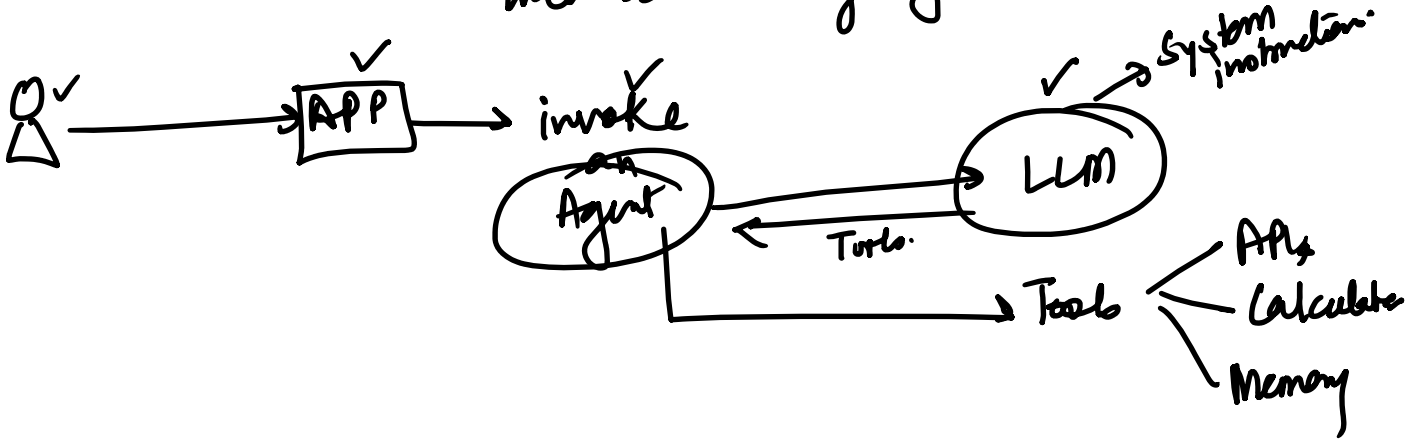
Q: Which pattern enables the entitlement lookup & support-case lookup to run together?

Ans: Parallelization





Agent Framework = foundation that helps us connect
[a model, instructions (System instructions), Tools (DB, Lambda, API calls, S3) & state]
into a working agent



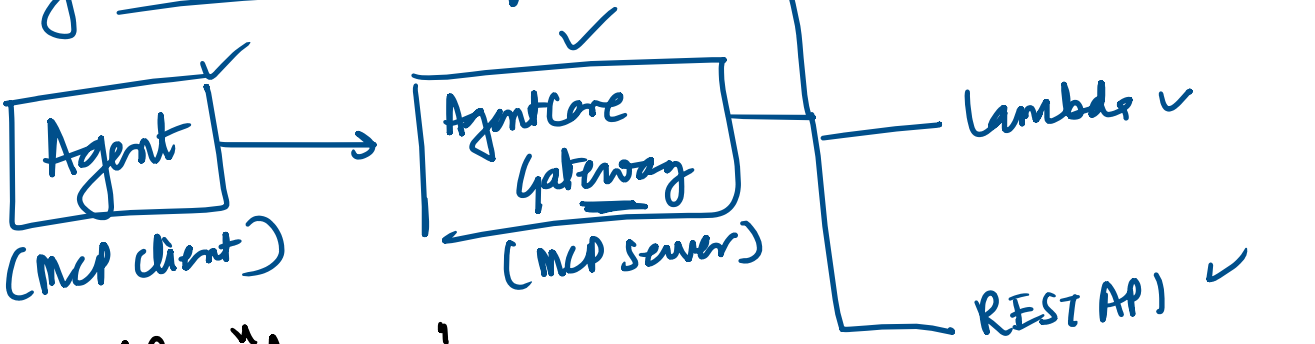
AgentCore = Managed Capabilities

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Disha
Is Agentic AI
- Runtime
- Gateway
- Identity
- Memory

1. Agent execution = Runtime
2. Tool integration = Gateway
3. Who ? what ? = Identity
4. Short-term & Long-term memory = Memory
5. Browser Automation
6. Code Execution

AgentCore Gateway



AgentCore "Memory"

	SHORT-Term	Long-term
Purpose	Maintain "active" conversational context	Preserve "selected info" across sessions
Typical Contents	Messages, current state, intermediate task context	user preferences, summaries, relevant facts.
Example	we are currently analyzing Company A for Q2.	The analyst prefers result in a Tabular format

A2A (Agent 2 Agent) vs. MCP (Model Context Protocol)

A2A (Agent 2 Agent) vs. MCP (Model Context Protocol)

- → agent Collaboration, distributed tasks, communicate among agents.
- → A2A = agent specialization

- → focus on capabilities of AI Application (LLM → external systems)
- ⇒ MCP = Reusable integrations to Tools & data

	A2A	MCP
Primary purpose	= Agent Collaboration	Tool & data integration
Main Interaction	= Agent → Agent	Host/Client ↔ MCP server
Discovery	= Agent Cards	Tools, Resources, prompt
Typical workflow	= Delegation & coordinated execution	Structured Tool/resources access
Enterprise Example	= Risk Agent delegates compliance review	Risk Agent retrieve financial data from RAG

Sample Agent Card :

```
{
 "name": "Financial Compliance Agent",
 "description": "Reviews financial findings against approved policies",
 "version": "1.0.0",
 "url": "https://example.com/a2a",
 "capabilities": {
  "streaming": true
 },
 "defaultInputModes": ["text", "application/json"],
 "defaultOutputModes": ["text", "application/json"],
 "skills": [
  {
   "id": "compliance_review",
   "name": "Compliance Review",
   "description": "Checks findings against approved rules",
   "tags": ["finance", "compliance"]
  }
 ]
```

```
}
```